Software Requirements Specification
for
Core Circuit Software's
Social Media Web Application

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1. Introduction

1.1 Purpose and Intended Audience

This document is produced, managed, and referenced by the members of Core Circuit Software. It aims to guide the development of a new online social media platform. It also serves as the agreement between Core Circuit Software and the Client, Edwin Rodríguez, regarding the specifications of the final product.

1.2 Product Scope

This social media platform aims to create an environment where users are free to easily share content between themselves. Whether content be about users personally, cover specific interests and/or hobbies, or simply act to host private conversations between users, this web app hopes to provide an easy to use platform for a variety of content.

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2. Overall Description

2.1 User Needs

As a social media platform for a variety of interests and hobbies, no specific demographics are being kept in mind. Despite this, a common trait that will be held by users would be the desire to share content with others as well as any silent observers scrolling through content for either research or simple entertainment. These traits will be present in most of the user base and must be kept in mind during development.

2.2 Assumptions and Dependencies

Amazon Web Services (AWS) will be used to host this application's database system with Django and Python being utilized to bridge the app to this database. Because of this, AWS and Django will act as cornerstones to this product. Development will also utilize REACT in the front end to accelerate User Interface creation to ultimately create an easy to use system for users to enjoy.

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3. System Features and Requirements

3.1 Functional Requirements

FR1. Account Creation

- FR1.1. Users can create accounts using a form that requires
  - FR1.1.1. A unique username
  - FR1.1.2. A password (minimum of 8 characters, not a common password or similar to the username)
  - FR1.1.3. An optional email address
- FR1.2. Upon form submission, users can optionally upload a profile picture and set a display name.

FR2. Account Login

- FR2.1. Existing users can log in using their username or email and password via the account login form.

FR3. Account Page

- FR3.1. Displays user information such as:
  - FR3.1.1. Profile picture
  - FR3.1.2. Username, display name, and bio
  - FR3.1.3. Number of friends and posts
  - FR3.1.4. Followed communities
- FR3.2. Users can access detailed lists of friends and communities by clicking on respective links
- FR3.3. Posts are sorted by date and can be viewed in detail by clicking on them

FR4. User Profile Customization

- FR4.1. Offer more options for users to customize their profiles, such as themes, background images

FR5. Friends

- FR5.1. Users can send and receive friend requests
- FR5.2. Friend lists are accessible through the user's profile, showing each friend's profile picture, name, and display name
- FR5.3. Users can send friend requests from within the friend list page and remove friends from their profiles

FR6. Private Messaging

- FR6.1. Users can send messages to each other

- FR6.2. The messaging page lists all conversations, showing the profile picture and display name of each participant
- FR6.3. Conversations can be muted or deleted, and users can view the online status of message recipients

FR7. Home feed

- FR7.1. The landing page after login, displaying posts from friends and followed communities

FR8. Post Creation

- FR8.1. Users can create posts via a form that includes fields for title, images, and description
- FR8.2. Users select whether to post to their profile or a community

FR9. Post Interaction

- FR9.1. Users can interact with posts by liking, disliking, sharing, or commenting
- FR9.2. Post interactions include icons for each action and display counters for likes, dislikes, and comments
- FR9.3. Comments can be made, edited, or deleted, and replies can be posted to other comments

FR10. Communities

- FR10.1. Users with common interests can join communities
- FR10.2. Community pages show details like name, icon, and number of followers
- FR10.3. Users can create communities by submitting a detailed form
- FR10.4. Communities feature post listings that users can sort and interact with similarly to personal posts

FR11. Search Functionality

- FR11.1. Implement a search bar to allow users to search for other users, posts, and communities by keywords

FR12. Security and Privacy Settings

- FR12.1. Allow users to customize their privacy settings, such as who can see their posts, profile information, and friend list.

3.2 Nonfunctional Requirements

1. **Security**

1. Security is one of the most important parts of our software, we take security of our user's data and safety seriously. Ensuring this safety will build trust.

NR1.1. Users will have limited attempts to login to their accounts, there will be a delay between each sign in attempt allowed then after many failed attempts the account will be locked and require password reset.

NR1.2. Users who forget their username or password can request to retrieve this information via a mechanism that verifies they are indeed the account owner

NR1.3. Users must be logged in to access the site, otherwise they will be redirected to the login page. Accounts will be remembered with the use of web tokens.

NR1.4. Passwords and other sensitive information must be stored securely on the database to ensure the safety of customer information.

NR1.5. User's actions will be logged for security purposes. This important changes such as login attempts and when the site was accessed.

NR1.6. Users will have control over the privacy of their accounts and posts, who can see and interact with them.

NR1.7. Users direct messages will be encrypted so even the user does not need to worry about snooping from networks

2. **Page design**

NR2.1. Page will consistently have a Left, Top, and Middle container whenever on the site

NR2.2. Top stretches across entire width, left sits below, and middle takes up everywhere else

NR2.3. Left container will contain navigation buttons to switch between:

- **2.3.1.** Home, Communities, Messages, and Friends
- **2.3.2.** Will also contain the "Create Post" button

NR2.4. Top container will contain:

- **2.4.1.** Logo in top left (navigates to home page)
- **2.4.2.** Search bar in the middle (to search for other users or communities)
- **2.4.3.** Profile picture in top right (navigates to profile page)

NR2.5. Middle container will display content determined by the left container
- **2.5.1.** Home will display posts made to friends' profiles (Default)
- **2.5.2.** Communities will display posts made in followed communities
- **2.5.3.** Messages will display all ongoing conversations, clicking on a specific conversation will expand it and show the conversation's history
- **2.5.4.** Friends will display all the user's friends, clicking on a friend will open that user's profile page
- **2.5.5.** Create Post will open the create post dialog, allowing user to set title and description, add media, and choose where to post (in x Community(ies), on Profile, or both)

3. Usability

NR3.1. The User should be able to access the instant message, profile page, Friends page, and popular page within 3 click of the home page.

NR3.2. The interface should be easy to learn and easy to find main actions such as making a post or editing their profile.

NR3.3. The interface should be simple and memorable enough for returning users to start working with it right away.

NR3.4. The interface should be clear and buttons should be distinct enough that users make very few mistakes.

4. Compatibility and Reliability

NR4.1. **Browser Compatibility**: The site must be accessible and fully functional across all modern web browsers, including but not limited to Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

NR4.2. **Data Consistency**: Data must be consistent and up-to-date across all instances. For example, if a user updates the text of their post, the changes should be immediately visible to all other users when the post is next loaded.

NR4.3. **Maintainability**: The site must be designed for easy updates and maintenance. This includes straightforward processes for updating features and fixing bugs.

NR4.4. **Scalability and modularity**: The architecture must support the seamless addition of new data types and features to the database and application without disrupting existing functionalities or requiring significant redesigns of the database structure.

NR4.5. **Uptime**: Aim for 99.9% uptime.

NR4.6. **Fault Tolerance**: Ensure the system can handle and recover from software or hardware failures without data loss.

5. **Performance**

NR5.1. **Response Time**: Ensure that the average response time for loading pages and processing requests is under 2 seconds.

NR5.2. **Capacity**: Support simultaneous access for at least 1,000 users without performance degradation.

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4. Agreement

Core System Software CEO,
Kevin Salcedo

Kevin Salcedo

Client,
Edwin Rodríguez

Edwin 09/20/2024